Direct vs. Indirect Cyclone Losses

Measurement, Geography, and Damage-Conditioned GDP Responses

1. Motivation & Data

Research Questions

1. How large are **direct physical losses** (property damage, deaths) relative to wind intensity, and how do they vary by storm category and country characteristics?
2. How does the **indirect GDP growth** response differ across cyclone intensity thresholds?
3. Does the **magnitude of direct damage** amplify the indirect GDP loss — i.e. does $\log(\text{damage}/\text{GDP}_{t-1})$ moderate the wind–GDP impulse response?

Data:

- EMDAT tropical cyclone damage & deaths, 1970–2015
- IBTrACS maximum wind speeds (m/s); PWT 9.0 macro panel
- 150+ countries; country-year panel

Econometric Specifications

Direct damage (static OLS with FE):

$$\log\left(\frac{\text{damage}_{it}}{\text{GDP}_{i,t-1}}\right) = \alpha + \beta\,\text{shock}_{it} + \mu_i + \delta_t + \varepsilon_{it}$$

Shock: continuous max wind (m/s) or category indicators (Cat 1 only: 33–43 m/s; Cat 2 only: 43–50 m/s; Cat 3+: $\geq$50 m/s). Country-clustered SE.

GDP local projection (LP):

$$\log y_{i,t+h} - \log y_{i,t-1} = \beta_h\, D_{it} + \gamma' \mathbf{X}_{it} + \mu_i^{(1)} t + \mu_i^{(2)} t^2 + \delta_{rt} + \varepsilon_{iht}$$

D_{it}: Cat 1+/2+/3+ binary. $\mathbf{X}_{it}$: two lags of shock + two lags of $\Delta \log y$. FE: country quadratic trends + region$\times$year. Driscoll–Kraay SE. $h = 0, \ldots, 8$.

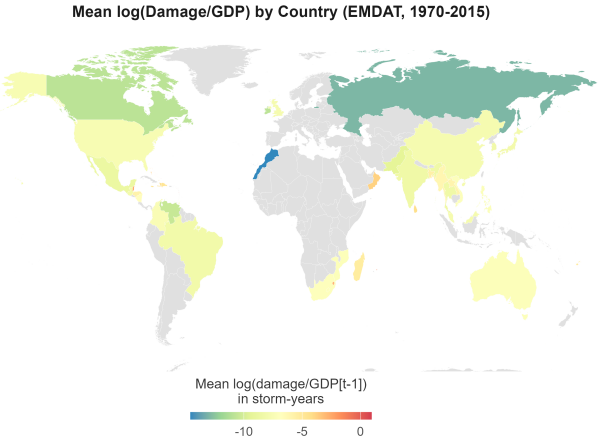
State-dependent LP:

$$\log y_{i,t+h} - \log y_{i,t-1} = \beta_h\, D_{it} + \gamma_h\, D_{it} \times \log\left(\frac{\text{damage}_{it}}{\text{GDP}_{i,t-1}}\right) + \cdots$$

Marginal effect of D_{it} evaluated at p_{10}, p_{50}, p_{90} of $\log(\text{damage}_t/\text{GDP}_{t-1})$; delta-method SE.

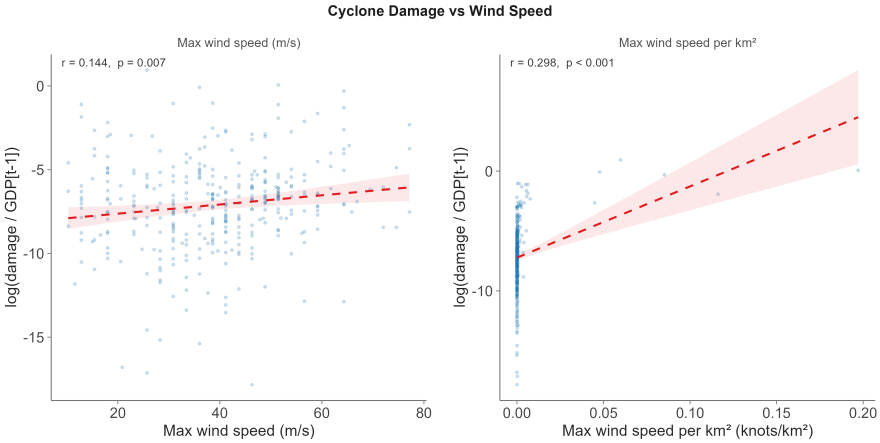
2. Direct Damage: Measurement

Geography of Direct Losses



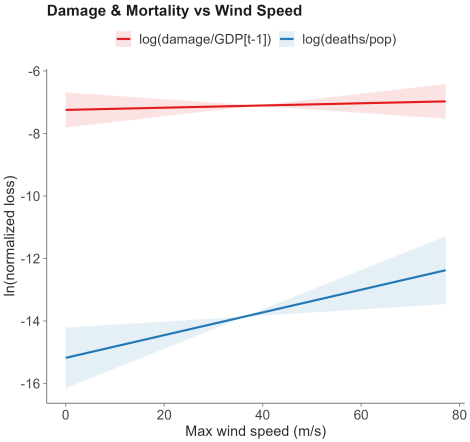
Mean log(EMDAT damage/GDP$_{t-1}$) across storm-years by country. Small island states and South/South-East Asia concentrate the highest normalised losses; North America and Europe show comparatively low ratios.

Damage Increases with Wind Speed



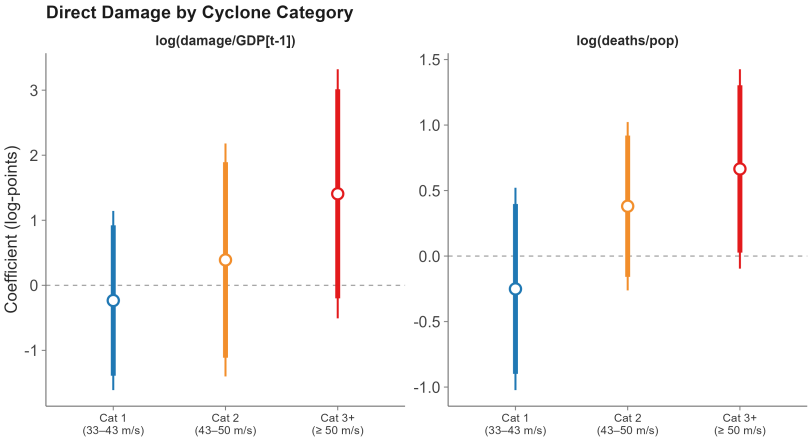
Left: max wind speed (m/s). Right: max wind per km^2 (knots/km^2). Both measures show a significant positive relationship with log(damage/GDP$_{t-1}$). Pearson r and p-value annotated; dashed line is the OLS fit.

Predicted Damage & Mortality vs Wind Speed



Predicted ln(normalised loss) as a function of max wind speed, OLS with country quadratic trends and region×year FE. Both log(damage/GDP$_{t-1}$) and log(deaths/pop) rise with wind, with mortality losses steeper at high intensities.

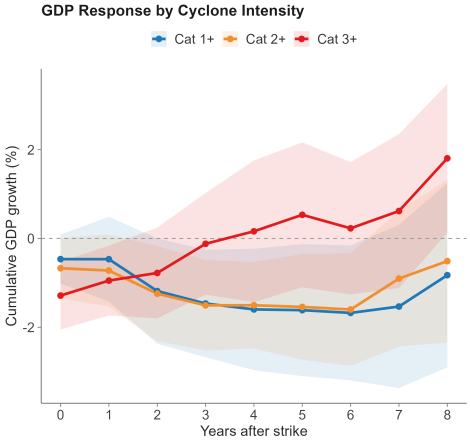
Category-Specific Damage Coefficients



Contemporaneous OLS coefficient (country-clustered SE) on three *mutually exclusive* category indicators. Left panel: $\log(\text{damage}/\text{GDP}_{t-1})$; right panel: $\log(\text{deaths}/\text{pop})$. Thick bar = 90 % CI; thin bar = 95 % CI. Both damage and mortality rise monotonically with storm intensity.

3. GDP Growth Response

GDP LP IRF by Cyclone Intensity



LP impulse responses of log GDP per capita to Cat 1+, Cat 2+, and Cat 3+ binary shocks. Cat 3+ strikes produce the largest and most persistent growth drag; Cat 1+ effects are smaller and recover faster. Shaded bands: 95 % Driscoll–Kraay confidence intervals.

4. Damage-Conditioned GDP Response

Does Higher Prior Damage Amplify GDP Losses?

If direct physical damage *mediates* the wind–GDP transmission:

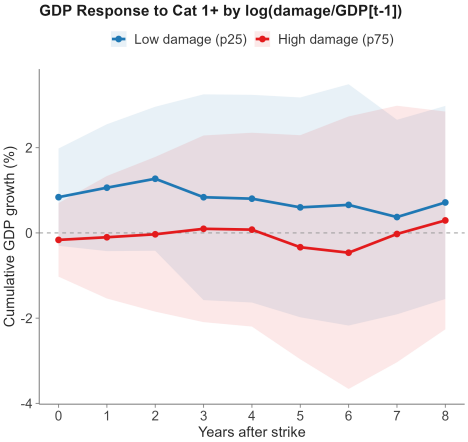
- Countries with larger $\log(\text{damage}/\text{GDP}_{t-1})$ in previous years are more vulnerable and should exhibit **deeper** GDP losses from the same cyclone shock
- The state-dependent LP isolates this channel, holding wind intensity (Cat 1+ indicator) constant

Specification:

$$\Delta_h \log y_{it} = \underbrace{\beta_h + \gamma_h \log\left(\frac{\text{damage}_{it}}{\text{GDP}_{it}}\right)}_{\text{ME}(D_{it}=1)} \times \mathbf{1}[\text{Cat 1+}]_{it} + \cdots$$

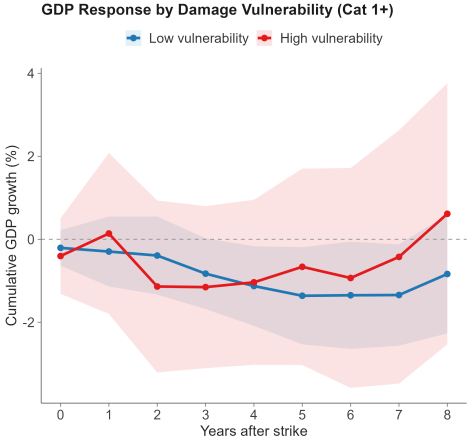
Marginal effect evaluated at p_{25} and p_{75} of contemporaneous $\log(\text{damage}/\text{GDP}_{t-1})$ via delta method.

State-Dependent LP: GDP Response Conditional on Damage Level



Countries at the **75th percentile** of prior log(damage/GDP$_{t-1}$) suffer larger and more persistent GDP losses from a Cat 1+ strike. Countries at the **25th percentile** show a muted response. Evaluation at p25/p75 matches the binary vulnerability split used in the sub-sample LP.

Sub-Sample LP: GDP IRF by Damage-Vulnerability (Binary)



Vulnerability index: country-mean of $\log(\text{damage}_t/\text{GDP}_{t-1})$ in storm-years with positive EMDAT damage. Countries split at the median into Low and High vulnerability; LP estimated separately within each sub-sample (all country-years). **High-vulnerability** countries show larger GDP losses, corroborating the state-dependent result with a fully separate estimation sample.

5. Reconciling the Two Approaches

Why Do the Two Approaches Differ?

Both approaches test damage heterogeneity but differ in design:

	State-dep LP	**Binary sub-sample**
Variation	Within-country	Between-country
State var	Contemporaneous D_t	Long-run mean $\bar{D}_i$
Sample	Damage-positive obs	All country-years
Functional form	Linear at p25/p75	Discrete binary split

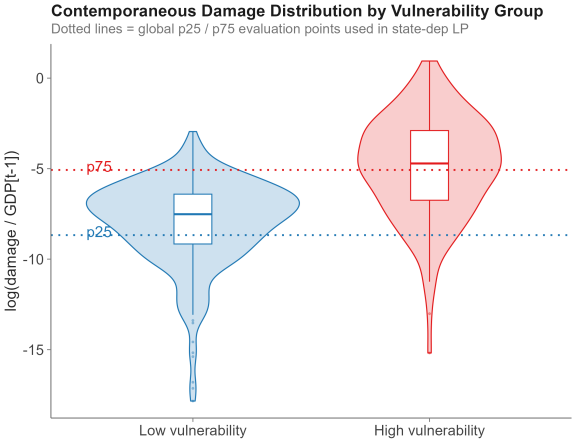
Three hypotheses:

H1 Baseline comparison Full-panel interaction LP (`lp_panel_inter`) as the discrete counterpart to the continuous state-dep LP.

H2 Within vs. between State-dep exploits annual fluctuations in damage; binary split exploits permanent country differences.

H3 Functional form Linear interaction at p25/p75 vs. discrete binary split.

EDA: Contemporaneous Damage by Vulnerability Group



Distribution of contemporaneous $\log(\text{damage}/\text{GDP}_{t-1})$ in storm-years with positive damage, by binary vulnerability group. Dotted lines mark the global p25/p75 evaluation points used in the state-dep LP. Overlap between groups indicates that the permanent vulnerability index and the contemporaneous state variable are measuring related but distinct objects.

Vulnerability Index Construction

Definition: for each country i, compute the mean of $\log(\text{damage}_{it}/\text{GDP}_{i,t-1})$ over all storm-years with positive EMDAT damage:

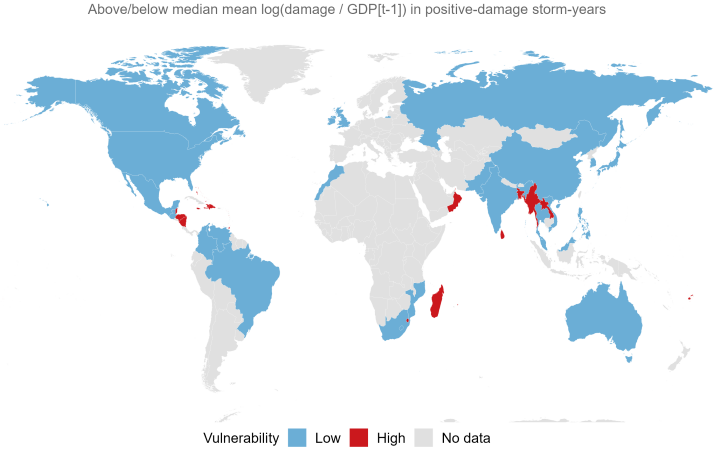
$$v_i = \frac{1}{|S_i|} \sum_{t \in S_i} \log\left(\frac{\text{damage}_{it}}{\text{GDP}_{i,t-1}}\right), \quad S_i = \{t : \text{maxwind}_{it} > 0,\ \text{damage}_{it} > 0\}$$

Countries are then split at the median of v_i into Low and High vulnerability. Countries with no positive-damage storm record receive no assignment (NA).

Why the log?

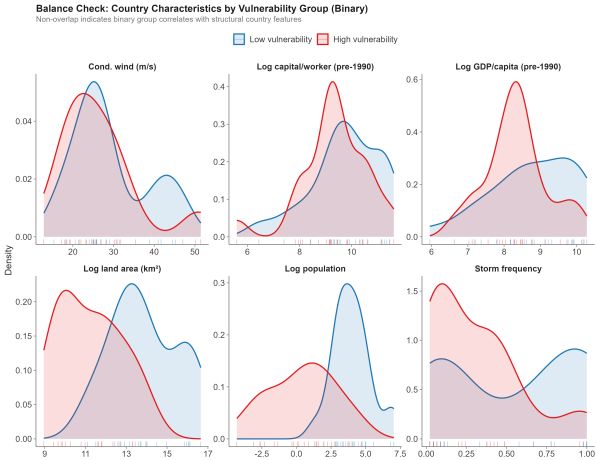
- Consistent with the state-dep LP state variable ($\log D_t/\text{GDP}_{t-1}$)
- Compresses extreme values; damage/GDP is heavily right-skewed
- Ensures the vulnerability index lives on the same scale as the interaction term

Map: Vulnerability Binary Split by Country



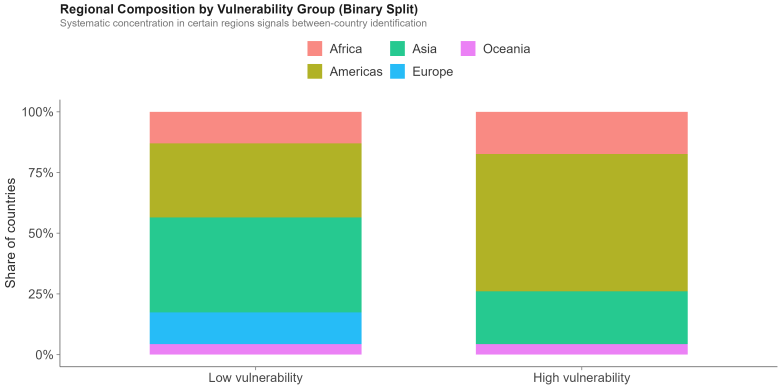
Countries shaded by binary vulnerability group (above/below median v_i). High-vulnerability countries (red) concentrate in South and South-East Asia, the Caribbean, and the Pacific. Grey = no positive-damage storm record in EMDAT, 1970–2015.

Balance Check: Is Tercile Assignment Quasi-Random?



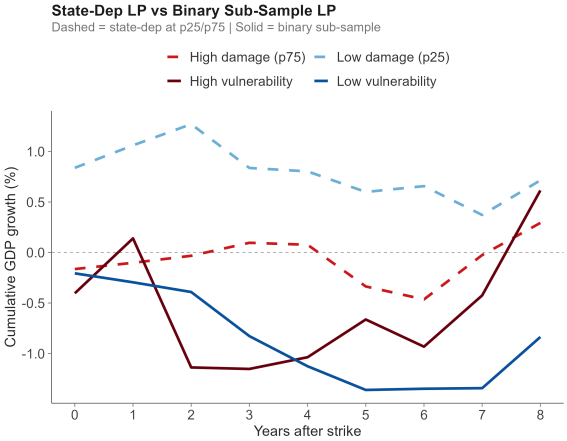
Kernel density of pre-determined country characteristics (GDP/capita, population, land area, storm frequency, conditional wind intensity, capital per worker) by binary group. Systematic separation indicates the binary split proxies structural features (income, geography, size) rather than pure damage vulnerability — relevant for interpreting the between-country identification in

EDA: Regional Composition by Vulnerability Group



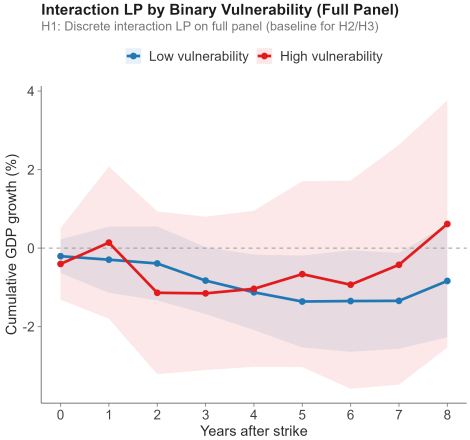
Share of countries by continent within each vulnerability group. Systematic concentration in Asia/Pacific for High vulnerability reflects the geography of cyclone exposure rather than pure damage vulnerability.

Direct Comparison: State-Dep vs Tercile LP



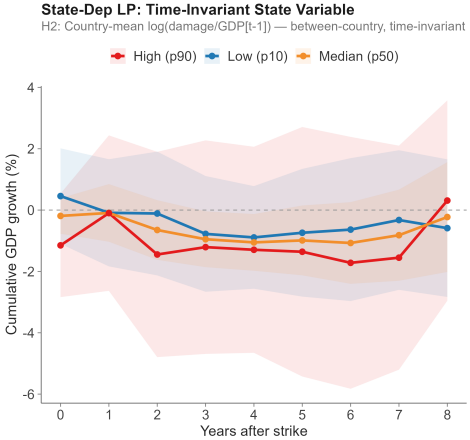
Dashed lines: state-dep LP evaluated at global p10/p50/p90 of contemporaneous $\log(D_t/\text{GDP}_{t-1})$. Solid lines: tercile sub-sample LP (all years within each tercile). Agreement in *ranking* with disagreement in *level* points to **H1** or **H2**; disagreement in ranking points to **H3**.

H1 – Storm-Years Only: Tercile LP



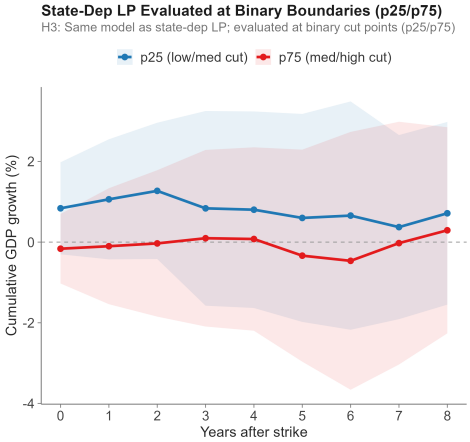
Full-panel interaction LP (`lp_panel_inter`) using `i(damage_binary, wind_cat1plus)`. Gives separate wind-on-GDP slopes for Low and High vulnerability directly from a single pooled regression. Serves as the discrete-interaction baseline for comparing with H2 (time-invariant state) and H3 (continuous state-dep at p25/p75).

H2 – Time-Invariant State Variable



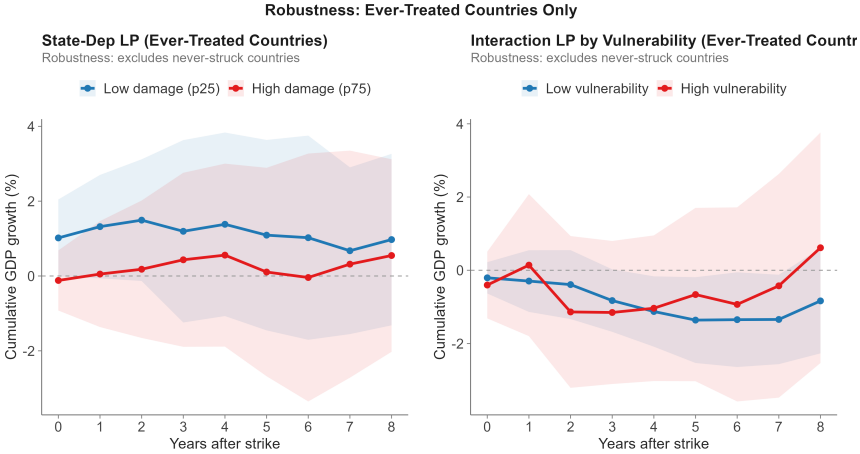
State-dep LP re-estimated replacing contemporaneous $\log(D_t/\mathrm{GDP}_{t-1})$ with each country's long-run mean over storm-years (time-invariant). A time-invariant state variable captures between-country differences, bringing the state-dep LP conceptually in line with the tercile approach. Convergence here isolates **H2** as a key driver.

H3 – State-Dep Evaluated at Tercile Boundaries



Same state-dep LP model; marginal effects evaluated at the tercile cut points p33 and p67 rather than p10/p90. If **H3** (functional form) matters, results at p33/p67 should bracket the tercile LP estimates more closely than the wide p10/p90 range.

Robustness: Ever-Treated Countries



Both heterogeneity approaches re-estimated on countries that were ever struck by any cyclone (maxwind > 0 in at least one year). Never-struck countries contribute only to fixed-effect precision in the full panel; excluding them tests whether they distort the heterogeneity estimates.

6. Takeaways

Summary

1. **Direct losses scale with wind.** Both log(damage/GDP$_{t-1}$) and log(deaths/pop) increase monotonically from Cat 1 to Cat 3+, with mortality losses proportionally steeper.

2. **GDP drag is intensity-graded.** Cat 3+ strikes produce deeper and more persistent GDP declines than Cat 1+ events; effects accumulate over 3–5 years.

3. **Prior damage amplifies the GDP channel.** The state-dependent LP shows that the same Cat 1+ shock generates substantially larger output losses when contemporaneous log(damage/GDP$_{t-1}$) is high, consistent with a damage–vulnerability trap.

4. **Heterogeneity matters.** Country damage vulnerability explains a significant share of cross-country variation in GDP responses, over and above wind intensity alone.

5. **Caveat.** EMDAT under-reports damages for small/poor countries; true direct losses — and the amplification channel — may be larger than estimated.